

Project work

The deadline for submitting the project report and material is decided on basis of this [poll](https://forms.office.com/Pages/ResponsePage.aspx?id=IXkdCjS3KU2LkLqAB1vdbab5PoK5LNtHhTxfzIYVrIUMk8wWVdUU0hYSE9ZWTI5RVFVODhLRkl3Ni4u)  (<https://forms.office.com/Pages/ResponsePage.aspx?id=IXkdCjS3KU2LkLqAB1vdbab5PoK5LNtHhTxfzIYVrIUMk8wWVdUU0hYSE9ZWTI5RVFVODhLRkl3Ni4u>). The project counts for 50% of the final grade in the course, and must be completed with a passing grade in order for you to be qualified to take the exam.

Goal of the project

The project work lets you dive deep into the engineering aspects of deep learning. It will give you a feel for how it is to develop deep learning-based solutions in real life, with end-to-end considerations of model selection, optimisation, evaluation, and development.

The end goal of the project is for you to **show what you have learned** in the course. This means to investigate problems that can be solved by deep learning, **design** and **train** a deep learning model, take relevant measures to make it perform optimally, and demonstrate its use on unseen data.

Project topic

You choose the project topic yourself, either based on the suggestions in the [project catalog](https://hvl.instructure.com/courses/28951/pages/project-catalog) (<https://hvl.instructure.com/courses/28951/pages/project-catalog>), or you can define it freely. In any case you should consider what problem you wish to solve, what data you will use, and how you expect deep learning to be useful. The topic must be approved beforehand by submitting a description in the [assignment](https://hvl.instructure.com/courses/32052/assignments/99106) (<https://hvl.instructure.com/courses/32052/assignments/99106>) here on Canvas.

Scope

There are two levels of expectations for the project work: Things that are *mandatory*, which you have to do in order to pass the course, and things that are *recommended*, which will improve the quality of your work.

The **mandatory** items you must complete are to

- Identify a use case for deep learning methods
- Construct and train a deep neural network
- Make an objective evaluation of model performance



We highlight that you must have designed and trained at least one model yourself -- see *Rules* below for more details concerning e.g pretrained models.

The **recommended** items are to

- Compare different strategies, such as different model architectures, optimisation algorithms, data preprocessing, feature engineering, and so on
- Make use of relevant tools for hyperparameter optimisation and performance monitoring
- Investigate quality of predictions by computing additional metrics and using tools for model explanation/interpretation

While an additional **highly recommended** item is to

- Deploy the model as a web app, where users can interact with it

For the recommended steps, you do not have to complete all, but you should focus on the ones that are relevant to your use case.

Feedback before the deadline

I am of course happy to help with defining the project and suggesting solutions. During the semester we will set aside time especially for this.

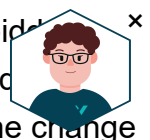
Requirements for the submission

For the actual submission, we require the following (both points are mandatory):

- A report that documents your work. This report must clearly state the objective, describe how you implemented the solution, and discuss the quality of the results. Templates for the report are provided for **Word (Norwegian)** (<https://hvl.instructure.com/courses/32052/files/3700418?wrap=1>)  (https://hvl.instructure.com/courses/32052/files/3700418/download?download_frd=1) , **Word (English)** (<https://hvl.instructure.com/courses/32052/files/3700417?wrap=1>)  (https://hvl.instructure.com/courses/32052/files/3700417/download?download_frd=1) , and **LaTeX**  (<https://www.overleaf.com/read/wmbdgkzfcscr#4f2577>) . You are free to choose if you write in Norwegian or English.
- Create a public GitHub repository which contains all the code, documents, and other material required for understanding and reproducing your results.

Additional rules and amendments

- You are allowed to use AI coding tools and take inspiration from solutions you find on the Internet, but you need to **cite sources** you use. Plagiarism is, as always, strictly forbidden.
- **You have to train at least one model from ground up** (i.e. from randomly initialised weights). If you take inspiration from an existing model, you have to make at least one change



to its architecture.

- As long as you have trained at least one model yourself, you can compare it to pretrained models, without having to re-train these other models as well.
- As long as you have trained at least one model yourself, you can extend your project with additional AI tools like OpenAI chat, DALL-E generated images, and other things that enhance your results.
- If you for some reason do not wish to make your repository public, you must add me (smaeland) as a collaborator. Remember that your GitHub profile serves as a nice portfolio to impress future employers.
- Other git services like GitLab are acceptable if the repository is public and all other requirements are satisfied.
- You can not re-use text and results you have or will submit for other graded work. So if you are working on deep learning for your bachelor or master project, be careful that there is no overlap with the DAT255 project. Ask if you are in doubt.
- You are free to use any frameworks you find necessary. However, since your implementation is part of the evaluation, all code must be written in Python.
- There are no strict requirements on how short/long the report should be, since well-made graphs and tables reduce the amount of text needed to document your work, and also we expect more from big groups than smaller ones. However, around 2500-3000 words is a good target.

Evaluation criteria

The project will be graded based on the following criteria:

1. General impression

To what extent does work exhibit proficiency in concepts and methods relevant to the course? Is the chosen project relevant to, and suitable for, solutions based on deep learning? Does the work show initiative in exploring suitable methods and studying relevant literature? To what extent does the work demonstrate both methodological and creative thinking?

2. Theoretical level

To what extent does the report document theoretical insight in to the chosen methods, and overview over potential methods? Are the chosen methods anchored to the contents of the course? Is relevant literature appropriately cited?

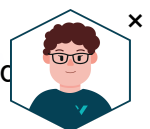
3. Implementation

Does the work document an ability to plan, structure and execute a project of the expected scope? Has necessary information, material and data been collected to ensure the quality of the implementation? Are experiments conducted in a structured and reproducible way? Are tools and methods

4. Results

4.1 Obtained results

Are the results presented in a structured and understandable way? Are the collected data sufficient to evaluate the outcome of the work?



4.2 Analysis

To what degree is the analysis and the discussion of results anchored to the contents of the course? Are results analysed in a methodological way?

4.3 Reflection

To what degree does the report offer reasonable evaluation of the results, and the implications of the results? Are the implications of the results discussed in context of comparable solutions and scientific work? Does the report document critical thinking around different sources of information? Are methodological and systematic sources of error considered and reflected upon? Does the conclusion describe to which degree the goals of the project are achieved? Are reasonable suggestions for improvements and further work given?

5. Presentation

To what extent does the report follow a logical structure? Is both the report, and the submitted code, presented in an understandable way? Are the contents of the report communicated with clear language at a suitable technical level? What is the quality of figures and illustrations?

